Practical No 1: Part B

Aim: Study of basic network command and Network configuration commands.

Theory:

Hostname

The HOSTNAME command displays the hostname of the system. The hostname command is much easier to use than going into the system settings to search for it.

C ls command is used to clear command prompt.

Ctrl + C is used to break the running command.

Ncpa.cpl: NCPA= Network Control Panel Applet, CPL= Control Panel.

ncpa. cpl is **a command that redirect the user to the network adaptors of the computer**

Ipconfig

Displays all current TCP/IP network configuration values and refreshes Dynamic Host Configuration Protocol (DHCP) and Domain Name System (DNS) settings. Used without parameters, ipconfig displays Internet Protocol version 4 (IPv4) and IPv6 addresses, subnet mask, and default gateway for all adapters.

Ipconfig/all

the /all parameter used with ipconfig will list all of your network adapters' configuration information. Note that this command will begin with the Windows IP Configuration section, which includes the following six items:

- Host Name
- Primary Dns Suffix
- Node Type
- IP Routing Enabled
- WINS Proxy Enabled
- DNS Suffix Search List

Ipconfig /displaydns

This /displaydns parameter shows the DNS resolver cache of your system. The cache cuts down on network traffic since it keeps track of IP addresses and website names you have already visited. As well as the contents (if any) of your local Hosts file. Each record contains the following sections:

- Each section begins with the **name of the cached domain** and then contains one or more records.
- **Record Name**: The name of the remote computer.
- **Record Type**: Common ones include 1 for type A (IP4 address), 28 for AAAA (IPv6), 15 for MX (Mail Exchange), 2 for NS (Name Server).
- **Time To Live**: displays when the entry will expire (in seconds).
- **Data Length**: The length in bytes (e.g., IP4 addresses have a length of four bytes, IPv6 has 16 bytes)
- **Section**: typically you'll find "Answer" listed in there, but it might also be "Additional."
- **A (Host) Record**: The IP address for the record.

Ipconfig /flushdns

The /flushdns parameter will flush the DNS resolver cache. This can be useful when you are troubleshooting or when you want to get rid of defective or obsolete DNS records. The cache will be repopulated as you browse the Internet or during normal system activity.

A reboot of your system also flushes the DNS resolver cache. This helps to keep some privacy, but it doesn't get rid of your browsing history.

Ping <<ipv4 address>>

The Ping command is one of the most widely used commands in the prompt tool, as it allows the user to check the connectivity of our system to another host.

This command sends four experimental packets to the destination host to check whether it receives them successfully, if so, then, we can communicate with the destination host. But in case the packets have not been received, that means, no communication can be established with the destination host.

Ping –t <<ip address>>: this command is used to pin continuously.

Pathping << IP address or anywebsite>>

This command sends multiple echo Request messages to each router between a source and destination, over a period of time, and then computes results based on the packets returned from each router.

TRACERT

The TRACERT command is used to trace the route during the transmission of the data packet over to the destination host and also provides us with the "hop" count during transmission.

Using the number of hops and the hop IP address, we can troubleshoot network issues and identify the point of the problem during the transmission of the data packet.

NETSTAT

The Netstat command as the name suggests displays an overview of all the network connections in the device. The table shows detail about the connection protocol, address, and the current state of the network.

ARP

The ARP command is used to access the mapping structure of IP addresses to the MAC address. This provides us with a better understanding of the transmission of packets in the network channel.

Systeminfo

Using the SYSTEMINFO command, we can access the system's hardware and software details, such as processor data, booting data, Windows version, etc.

nslookup: Displays information from Domain Name System (DNS) name servers.

NOTE :If you write the command as above it shows as default your pc's server name firstly.

Screenshot:

```
C:\WINDOWS\system32\cmd.

Microsoft Windows [Version 10.0.26200.7462]
(c) Microsoft Corporation. All rights reserved.

C:\Users\niraj>hostname
Niraj

C:\Users\niraj>
```

```
C:\WINDOWS\system32\cmd.

Windows IP Configuration


Wireless LAN adapter Local Area Connection* 1:

   Media State . . . . . . . . . . . : Media disconnected
   Connection-specific DNS Suffix  . :

Wireless LAN adapter Local Area Connection* 2:

   Media State . . . . . . . . . . . : Media disconnected
   Connection-specific DNS Suffix  . :

Wireless LAN adapter Wi-Fi:

   Connection-specific DNS Suffix  . :
   IPv6 Address. . . . . . . . . . . : 2402:8100:3135:7146:1256:9ed1:55aa:e1bc
   Temporary IPv6 Address. . . . . . : 2402:8100:3135:7146:1193:cf57:f7b:10da
   Link-local IPv6 Address . . . . . : fe80::240d:b2cd:a0be:d24%4
   IPv4 Address. . . . . . . . . . . : 172.19.224.76
   Subnet Mask . . . . . . . . . . . : 255.255.255.0
   Default Gateway . . . . . . . . . : fe80::a42b:8fff:fe19:98fa%4
                                       172.19.224.75

Ethernet adapter Ethernet:

   Media State . . . . . . . . . . . : Media disconnected
   Connection-specific DNS Suffix  . :

C:\Users\niraj>
```

```
C:\WINDOWS\system32\cmd.

C:\Users\niraj>ipconfig/displaydns

Windows IP Configuration

    fe3cr.delivery.mp.microsoft.com
    ----------------------------------------
    Record Name . . . . . : fe3cr.delivery.mp.microsoft.com
    Record Type . . . . . : 5
    Time To Live  . . . . : 598939
    Data Length . . . . . : 8
    Section . . . . . . . : Answer
    CNAME Record  . . . . : fe3.delivery.mp.microsoft.com


    Record Name . . . . . : fe3.delivery.mp.microsoft.com
    Record Type . . . . . : 5
    Time To Live  . . . . : 598939
    Data Length . . . . . : 8
    Section . . . . . . . : Answer
    CNAME Record  . . . . : glb.cws.prod.dcat.dsp.trafficmanager.net


    Record Name . . . . . : glb.cws.prod.dcat.dsp.trafficmanager.net
    Record Type . . . . . : 28
    Time To Live  . . . . : 598939
    Data Length . . . . . : 16
    Section . . . . . . . : Answer
    AAAA Record . . . . . : 2a01:111:f100:a000::4134:4847
```

```
C:\Users\niraj>ipconfig /flushdns

Windows IP Configuration

Successfully flushed the DNS Resolver Cache.

C:\Users\niraj>
```

```
C:\WINDOWS\system32\cmd.

C:\Users\niraj>ipconfig/displaydns

Windows IP Configuration

    79.137.168.92.in-addr.arpa.
    ----------------------------------------
    Record Name . . . . . : 79.137.168.92.in-addr.arpa.
    Record Type . . . . . : 12
    Time To Live  . . . . : 604800
    Data Length . . . . . : 8
    Section . . . . . . . : Answer
    PTR Record  . . . . . : Redmi-10-Prime.mshome.net


    kubernetes.docker.internal
    ----------------------------------------
    No records of type AAAA


    Record Name . . . . . : kubernetes.docker.internal
    Record Type . . . . . : 1
    Time To Live  . . . . : 604800
    Data Length . . . . . : 4
    Section . . . . . . . : Answer
    A (Host) Record . . . : 127.0.0.1


    host.docker.internal
    ----------------------------------------
```

```
C:\WINDOWS\system32\cmd.

C:\Users\niraj>ipconfig

Windows IP Configuration


Wireless LAN adapter Local Area Connection* 1:

   Media State . . . . . . . . . . . . : Media disconnected
   Connection-specific DNS Suffix  . :

Wireless LAN adapter Local Area Connection* 2:

   Media State . . . . . . . . . . . . : Media disconnected
   Connection-specific DNS Suffix  . :

Wireless LAN adapter Wi-Fi:

   Connection-specific DNS Suffix  . :
   IPv6 Address. . . . . . . . . . . : 2402:8100:3135:7146:1256:9ed1:55aa:e1bc
   Temporary IPv6 Address. . . . . . : 2402:8100:3135:7146:1193:cf57:f7b:10da
   Link-local IPv6 Address . . . . . : fe80::240d:b2cd:a0be:d24%4
   IPv4 Address. . . . . . . . . . . : 172.19.224.76
   Subnet Mask . . . . . . . . . . . : 255.255.255.0
   Default Gateway . . . . . . . . . : fe80::a42b:8fff:fe19:98fa%4
                                       172.19.224.75

Ethernet adapter Ethernet:

   Media State . . . . . . . . . . . : Media disconnected
```

```
C:\WINDOWS\system32\cmd.

C:\Users\niraj>ping

Usage: ping [-t] [-a] [-n count] [-l size] [-f] [-i TTL] [-v TOS]
            [-r count] [-s count] [[-j host-list] | [-k host-list]]
            [-w timeout] [-R] [-S srcaddr] [-c compartment] [-p]
            [-4] [-6] target_name

Options:
    -t             Ping the specified host until stopped.
                   To see statistics and continue - type Control-Break;
                   To stop - type Control-C.
    -a             Resolve addresses to hostnames.
    -n count       Number of echo requests to send.
    -l size        Send buffer size.
    -f             Set Don't Fragment flag in packet (IPv4-only).
    -i TTL         Time To Live.
    -v TOS         Type Of Service (IPv4-only. This setting has been deprecated
                   and has no effect on the type of service field in the IP
                   Header).
    -r count       Record route for count hops (IPv4-only).
    -s count       Timestamp for count hops (IPv4-only).
    -j host-list   Loose source route along host-list (IPv4-only).
    -k host-list   Strict source route along host-list (IPv4-only).
    -w timeout     Timeout in milliseconds to wait for each reply.
    -R             Use routing header to test reverse route also (IPv6-only).
                   Per RFC 5095 the use of this routing header has been
                   deprecated. Some systems may drop echo requests if
                   this header is used.
    -S srcaddr     Source address to use.
    -c compartment Routing compartment identifier.
```

```
C:\Users\niraj>ping 2402:8100:3135:7146:1256:9ed1:55aa:e1bc

Pinging 2402:8100:3135:7146:1256:9ed1:55aa:e1bc with 32 bytes of data:
Reply from 2402:8100:3135:7146:1256:9ed1:55aa:e1bc: time<1ms
Reply from 2402:8100:3135:7146:1256:9ed1:55aa:e1bc: time<1ms
Reply from 2402:8100:3135:7146:1256:9ed1:55aa:e1bc: time<1ms
Reply from 2402:8100:3135:7146:1256:9ed1:55aa:e1bc: time<1ms

Ping statistics for 2402:8100:3135:7146:1256:9ed1:55aa:e1bc:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 0ms, Average = 0ms

C:\Users\niraj>
```

```
C:\Users\niraj>pathping 2402:8100:3135:7146:1256:9ed1:55aa:e1bc

Tracing route to Niraj [2402:8100:3135:7146:1256:9ed1:55aa:e1bc]
over a maximum of 30 hops:
  0  Niraj [2402:8100:3135:7146:1256:9ed1:55aa:e1bc]
  1  Niraj [2402:8100:3135:7146:1256:9ed1:55aa:e1bc]

Computing statistics for 25 seconds...
            Source to Here   This Node/Link
Hop  RTT    Lost/Sent = Pct  Lost/Sent = Pct  Address
  0                                          Niraj [2402:8100:3135:7146:1256:9ed1:55aa:e1bc]
                                 0/ 100 =  0%   |
  1    0ms     0/ 100 =  0%     0/ 100 =  0%  Niraj [2402:8100:3135:7146:1256:9ed1:55aa:e1bc]

Trace complete.

C:\Users\niraj>
```

```
C:\Users\niraj>tracert

Usage: tracert [-d] [-h maximum_hops] [-j host-list] [-w timeout]
               [-R] [-S srcaddr] [-4] [-6] target_name

Options:
    -d                 Do not resolve addresses to hostnames.
    -h maximum_hops    Maximum number of hops to search for target.
    -j host-list       Loose source route along host-list (IPv4-only).
    -w timeout         Wait timeout milliseconds for each reply.
    -R                 Trace round-trip path (IPv6-only).
    -S srcaddr         Source address to use (IPv6-only).
    -4                 Force using IPv4.
    -6                 Force using IPv6.

C:\Users\niraj>
```

```
C:\Users\niraj>tracert www.google.com

Tracing route to www.google.com [2404:6800:4002:81d::2004]
over a maximum of 30 hops:

  1     3 ms     3 ms     3 ms  2402:8100:3135:7146::50
  2     *        *        *     Request timed out.
  3    51 ms    20 ms    20 ms  2402:8100:17:160::1
  4    46 ms    31 ms    60 ms  fd00:0:16:16::6a
  5    73 ms    28 ms    34 ms  fd00:0:17:17::69
  6    72 ms    17 ms    23 ms  2400:c700:0:24::134
  7     *        *        *     Request timed out.
  8   127 ms    36 ms    35 ms  2402:8100:4000::68c
  9    71 ms    38 ms    32 ms  2402:8100:4000::959
 10    74 ms    42 ms    40 ms  2402:8100:4000::95b
 11   363 ms    45 ms    36 ms  2400:5200:401:a::1f
 12    66 ms    44 ms    31 ms  2001:4860:1:1::1706
 13    68 ms     *        *     2001:4860:0:1::7973
 14    39 ms    37 ms    34 ms  2001:4860:0:1::7974
 15    85 ms    38 ms    33 ms  2001:4860::c:4004:53ca
 16    84 ms    57 ms    59 ms  2001:4860::c:4004:153f
 17    68 ms    62 ms    59 ms  2001:4860::9:4001:d9e7
 18   138 ms    59 ms    63 ms  del11s18-in-x04.1e100.net [2404:6800:4002:81d::2004]

Trace complete.

C:\Users\niraj>
```

```
C:\Users\niraj>netstat

Active Connections

  Proto  Local Address          Foreign Address        State
  TCP    127.0.0.1:49671        kubernetes:49672       ESTABLISHED
  TCP    127.0.0.1:49672        kubernetes:49671       ESTABLISHED
  TCP    127.0.0.1:49673        kubernetes:49674       ESTABLISHED
  TCP    127.0.0.1:49674        kubernetes:49673       ESTABLISHED
  TCP    [2402:8100:3135:7146:1193:cf57:f7b:10da]:54416  sa-in-f188:https       ESTABLISHED
  TCP    [2402:8100:3135:7146:1193:cf57:f7b:10da]:54878  whatsapp-cdn6-shv-02-bom5:https  ESTABLISHED
  TCP    [2402:8100:3135:7146:1193:cf57:f7b:10da]:61072  [64:ff9b::acbc:9b19]:https  ESTABLISHED
  TCP    [2402:8100:3135:7146:1193:cf57:f7b:10da]:61076  [64:ff9b::142a:4158]:https  TIME_WAIT
  TCP    [2402:8100:3135:7146:1193:cf57:f7b:10da]:61574  [2603:1040:a06:6::1]:https  ESTABLISHED
  TCP    [2402:8100:3135:7146:1193:cf57:f7b:10da]:61575  [2603:1040:a06:6::1]:https  ESTABLISHED
  TCP    [2402:8100:3135:7146:1193:cf57:f7b:10da]:62869  [2a06:98c1:3107::6812:2715]:https  ESTABLISHED

C:\Users\niraj>
```

```
C:\WINDOWS\system32\cmd.

C:\Users\niraj>arp

Displays and modifies the IP-to-Physical address translation tables used by
address resolution protocol (ARP).

ARP -s inet_addr eth_addr [if_addr]
ARP -d inet_addr [if_addr]
ARP -a [inet_addr] [-N if_addr] [-v]

  -a            Displays current ARP entries by interrogating the current
                protocol data.  If inet_addr is specified, the IP and Physical
                addresses for only the specified computer are displayed.  If
                more than one network interface uses ARP, entries for each ARP
                table are displayed.
  -g            Same as -a.
  -v            Displays current ARP entries in verbose mode.  All invalid
                entries and entries on the loop-back interface will be shown.
  inet_addr     Specifies an internet address.
  -N if_addr    Displays the ARP entries for the network interface specified
                by if_addr.
  -d            Deletes the host specified by inet_addr. inet_addr may be
                wildcarded with * to delete all hosts.
  -s            Adds the host and associates the Internet address inet_addr
                with the Physical address eth_addr.  The Physical address is
                given as 6 hexadecimal bytes separated by hyphens. The entry
                is permanent.
  eth_addr      Specifies a physical address.
  if_addr       If present, this specifies the Internet address of the
                interface whose address translation table should be modified.
```

```
C:\WINDOWS\system32\cmd.

  > arp -a                                  .... Displays the arp table.

C:\Users\niraj>systeminfo

Host Name:                    NIRAJ
OS Name:                      Microsoft Windows 11 Home Single Language
OS Version:                   10.0.26200 N/A Build 26200
OS Manufacturer:              Microsoft Corporation
OS Configuration:             Standalone Workstation
OS Build Type:                Multiprocessor Free
Registered Owner:             nirajfating8554@gmail.com
Registered Organization:      HP
Product ID:                   00356-24691-92299-AAOEM
Original Install Date:        27-12-2024, 10:53:01
System Boot Time:             05-01-2026, 21:41:26
System Manufacturer:          HP
System Model:                 Victus by HP Gaming Laptop 15-fb0xxx
System Type:                  x64-based PC
Processor(s):                 1 Processor(s) Installed.
                              [01]: AMD64 Family 25 Model 80 Stepping 0 AuthenticAMD ~1908 Mhz
BIOS Version:                 AMI F.22, 23-07-2024
Windows Directory:            C:\WINDOWS
System Directory:             C:\WINDOWS\system32
Boot Device:                  \Device\HarddiskVolume1
System Locale:                en-us;English (United States)
Input Locale:                 00004009
Time Zone:                    (UTC+05:30) Chennai, Kolkata, Mumbai, New Delhi
Total Physical Memory:        7,518 MB
Available Physical Memory:    1,320 MB
Virtual Memory: Max Size:     19,315 MB
```

```
C:\Users\niraj>nslookup
Default Server:  UnKnown
Address:  172.19.224.75

>
```

Conclusion: Basic Network Configuration commands are studied and executed successfully.

Attendance	Performance	Viva	Total
3	7	5	15